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SmartGlycoAI

Stage 2: User and Task Analysis



Authors:

77314, Afonso Soeiro
77307, Martim Alcobia Nunes
76831, Matteo Philippe Demetri
77481, Rodrigo Teixeira Alves Silva Simões

Lab class Nº P4

Group Nº 6

Professor:
Teresa Romão

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SmartGlycoAI

Stage 2 - User and Task Analysis

1. Problem

People with type 1 diabetes do not produce insulin, so they have to replace it with injections or an insulin pump for the rest of their lives. This means that many of the decisions a healthy body makes automatically have to be made by the patient, several times a day, every day.

Counting carbohydrates and calculating doses

Before every meal, the patient has to estimate how many grams of carbohydrates are on their plate and then calculate the insulin dose using their personal settings: the insulin-to-carb ratio (how many grams one unit of insulin covers), the sensitivity factor (how much one unit lowers glucose) and the glycemia target. If glucose is already high before the meal, a correction has to be added to the calculation.
Estimating carbohydrates by eye is difficult. It is easier with packaged food that has a label, but most meals are not like that: a plate of rice with meat at a canteen, a dish cooked by someone else, or a meal at a restaurant. Portion sizes are hard to judge and mixed dishes hide ingredients such as sauces or bread. A small error in the estimate leads to a wrong dose, which can cause high glucose a few hours later or, worse, a low. Doing this calculation 3-5 times a day, often in public and in a hurry, is tiring and stressful, and many patients end up guessing or always eating the same ‘safe’ meals.

Risk of hypoglycemia

When there is too much insulin compared with the food eaten, or after physical activity, glucose can drop below the safe range. A low (hypoglycemia) causes shaking, sweating, confusion and, if it is not treated in time, can lead to fainting. Patients have to react quickly by eating fast-acting carbohydrates. The problem is that a low is often noticed only when the symptoms appear, for example on public transport, while driving or during sleep. Glucose sensors can show the current value, but the patient has to keep checking it and interpret the trend by themselves.

Scattered information

The information a patient needs is spread across different places: glucose values in the sensor app, the carbohydrates and doses in their memory, in notes on the phone or in a paper notebook, and the treatment settings on a sheet given by the doctor. Because of this, it is hard for the patient to understand how food, insulin and activity affect their glucose over the days, and mistakes in past records are rarely corrected.

Communication with the healthcare team

Endocrinologists and diabetes nurses see each patient only every few months, in appointments of about 15 to 20 minutes. To adjust the treatment, they need to know what happened between appointments. When glucose was high or low, what the patient ate and how much insulin they took. Today they usually get a sensor report and depend on what the patient remembers about meals and doses, so the picture is incomplete and the time of the appointment is spent collecting information instead of making decisions.

Our proposal

SmartGlycoAI is a mobile app that brings these tasks together. It shows the current glucose and recent trend from the CGM sensor, estimates the carbohydrates of a meal from a photo and suggests an insulin dose based on the user's own settings, warns the user before a predicted low with a suggestion of what to eat, and keeps a diary of meals and corrections that can be reviewed

and edited. Healthcare professionals access the data that patients share with them through a web version of the platform, so they can follow their patients between appointments.

2. Users

We identified three user classes: patients, endocrinologists and nurses. For each class we describe its main characteristics and a persona based on the person we interviewed (see Appendix 1).

2.1 Patients

Who	People with type 1 diabetes treated with insulin, from newly diagnosed to patients with many years of experience.
Age	Wide range, from teenagers to older adults (about 14-75).
Technology	Varies a lot. Younger users are used to apps and expect quick access, for example a widget on the phone's home screen. Older users may need simple screens, large text and clear feedback. Many use a CGM sensor (e.g. FreeStyle Libre, Dexcom).
Context and frequency	Everywhere (home, work, university, restaurants), several times a day. Usually quickly, before eating, often with people around them, so the app must be fast and discreet.

Persona: João Antunes - young adult with type 1 diabetes
'It's a continuous process that requires a lot of attention before every meal.'

Goals:

- See the current glucose and the graph of the last days in the first few seconds.
- Calculate carbohydrates from a photo of the plate, quickly and discreetly.
- Get reminders to check glucose and take insulin, and see clearly if his condition is improving.
- Receive AI recommendations about food, exercise and habits, as long as they are checked and validated by his healthcare team.

Frustrations: Remembering to check glucose, take insulin and keep healthy eating habits every day; the constant attention needed before every meal.

2.2 Endocrinologists

Who	Doctors specialised in diabetes who define and adjust the patient's treatment (ratios, basal doses, targets).
Age	About 30-65.
Technology	Used to computers and clinical software, but have little time for each patient. Prefer to analyse data on the office computer screen.
Context and frequency	In the office, before or during appointments. Each patient every few months; several patients a week.

Persona: Joana Silvia - endocrinologist
'AI should support medical decisions rather than replace professional judgment.'

Goals:

- Access glucose measurements, insulin history and health progress in one place
- See glucose history graphs and alerts for concerning patterns
- Understand the impact of meals throughout the week.

Frustrations: Consultations are less efficient when the data is not organised in one place, does not trust AI insights without accurate and secure data, clear explanations and transparency about how the algorithm was trained.

2.3 Nurses

Who	Nurses who follow diabetic patients between appointments and teach them how to manage their treatment, especially newly diagnosed and older patients.
Age	About 25-60.
Technology	Comfortable with apps and recommend them to patients, so they need to understand them well enough to explain them.
Context and frequency	Follow-up at the hospital or health centre and between appointments, several times a week.

Persona: Martim Garcia - nurse
'Some patients might have difficulties understanding medical information or navigating complicated interfaces, particularly older users.'

Goals:
- Monitor patients' glucose readings and care routines
- Identify patients who need additional support
- Help patients manage diabetes more independently.

Frustrations: Complicated interfaces and medical terms that patients do not understand (suggests a glossary, large buttons and simple graphs), notifications that are intrusive at work (suggests configurable notifications, such as a subtle vibration).

3. Tasks

We identified four high-level tasks. Tasks 1, 3 and 4 are performed by patients and Task 2 by healthcare professionals.

Task 1: Log a meal and get insulin recommendation (Patient)

Objective: Accurately log food intake using computer vision to receive an insulin dose suggestion.
Pre-conditions: User is logged in, the treatment settings (insulin-to-carb ratio, sensitivity factor and glycemia target) are filled in, and the app is connected to the CGM sensor.
Time/Freq: 3-5 times a day (before meals), under 1 minute.
Exceptions: Camera fails to recognize food; CGM sensor loses Bluetooth connection.
Sub-tasks:
1. Open the meal logging module.
2. Take a photo of the meal.
3. Confirm or adjust the AI's carbohydrate estimation.
4. Review and accept/reject the suggested insulin dose.

Task 2: Review patient's glycemic history (Doctor/Nurse)

Objective: Analyze weekly/monthly glucose trends to adjust medical treatment.
Pre-conditions: The patient has shared their data with the professional, who is logged in to the web version of the platform.
Time/Freq: Before or during each appointment with the patient (every few months), and occasionally between appointments to follow patients who need support.
Exceptions: Patient has not synced their data recently.
Sub-tasks:

1. Access the specific patient's profile.
2. Filter data by the last 30 days.
3. Identify patterns of hypoglycemia or hyperglycemia.

Task 3: Respond to a predictive low-glucose alert (Patient)

Objective: Act before a hypoglycemia happens, after the app predicts a fast drop in glucose.
Pre-conditions: CGM sensor is connected and notifications are allowed.
Time/Freq: A few times a week, whenever a fast drop is detected. Urgent: the user has about 20 minutes to act.
Exceptions: User does not see the notification (phone silent or in pocket); false alarm caused by a wrong sensor reading; user ignores the alert by mistake.
Sub-tasks:

1. Receive the alert notification with the predicted value and time.
2. Open the alert and read the suggestion (e.g. eat 15 g of fast-acting carbs).
3. Eat the suggested carbohydrates.
4. Register the correction in the diary, or ignore the alert if not needed.

Task 4: Review and edit the diary history (Patient)

Objective: Check past meals and corrections, and fix any entry with wrong values.
Pre-conditions: User is logged in and has at least one entry in the diary.
Time/Freq: Once a day or a few times a week (e.g. end of the day or before an appointment). About 1 to 2 minutes.
Exceptions: User deletes an entry by mistake; user enters an impossible value (e.g. 300 units of insulin); changes are not saved because the user leaves the screen.
Sub-tasks:

1. Open the Diary tab.
2. Scroll through the list of meals and corrections.
3. Open an entry to see its details (photo, time, carbs, insulin dose).
4. Change the time, description, carbs or insulin dose.
5. Save the changes, or delete the entry.

4. Scenarios

Scenario 1 - Lunch at a restaurant (Task 1)

João Antunes is a young adult with type 1 diabetes who uses a FreeStyle Libre sensor. On Tuesday at 1 p.m., he is having lunch with his friends at a restaurant and orders steak with rice and salad. When the food arrives, his glucose is 115 mg/dL and stable. He needs to figure out the carbohydrates in his meal, which he suspects has more rice than usual (around 60g instead of an initial guess of 45g). Using the system, his goal is to register this meal discreetly, adjust the carbohydrate estimation, and receive the correct insulin dose recommendation before he starts eating, so his friends barely notice.

Scenario 2 - A low on the way home (Tasks 3 and 4)

At 6 p.m., João is on the bus home after the gym. He receives a predictive alert indicating his glucose will reach 65 mg/dL in 20 minutes and suggesting he consume 15 g of fast-acting carbohydrates. He drinks a small pack of juice. He must use the system to log that he took this corrective action. Later that evening, at home, he reviews his day. He remembers that at lunch he left part of the rice on the plate, meaning he ate 45 g instead of the 60 g he previously recorded. His goal now is to use the system to locate his past lunch entry and correct the carbohydrate amount, ensuring his daily logs are accurate.

Scenario 3 - Preparing an appointment (Task 2)

Dr. Joana Silvia is an endocrinologist at a hospital in Almada, and João has an appointment with her on Thursday. Before the consultation, she uses the clinical system on her office computer to review João's data from the past 30 days. She needs to identify any problematic patterns, such as the late afternoon lows on gym days and high glucose after lunch. During the appointment, they discuss the data and decide to change his insulin-to-carb ratio from 1 unit per 15 g to 1 unit per 12 g. João must then use the system to update this clinical setting. In the following weeks, Nurse Martim Garcia will use the platform to monitor João's glucose readings remotely to see how he is adapting.

5. Interviews

ID	User class	Profile	How / where
P1	Patient	João Antunes, young adult, type 1 diabetes	In person
D1	Endocrinologist	Joana Silvia	In person
N1	Nurse	Martim Garcia	In person

Summary of results

We interviewed three participants, one from each user class: a patient (P1), an endocrinologist (D1) and a nurse (N1). The main conclusions and their impact on the design are the following:

1. **Quick access to the current glucose.** João wants to see his current glucose and the graph of the last days in the first few seconds, especially in an emergency, and suggested a widget on the phone's home screen. This supports showing the current value as the first and largest element of the home screen. A widget will be considered for a later version.
2. **Photo carb counting is accepted if it is fast and discreet.** João would use the camera in a restaurant as long as the process is quick. This confirms Task 1 as the main task and that it should take less than a minute, with as few steps as possible.
3. **AI should support, not replace, decisions.** João wants AI advice validated by his healthcare team, and Dr. Silvia said that AI should support medical decisions rather than replace professional judgment, with clear explanations of its insights. This supports letting the user always review and adjust the carbohydrates and the dose before saving (Task 1), showing how the dose was calculated, and including information about how the algorithm works.
4. **Professionals need all the data in one place.** Dr. Silvia wants glucose, insulin history and progress in a single place and prefers to analyse it on her office computer. Martim wants to easily identify patients who need support. This supports a web version for professionals with glucose history graphs and alerts for concerning patterns (Task 2).
5. **Reminders are important for patients**. Both João and Martim mentioned reminders to check glucose and take insulin, and Martim also suggested notifications about missed measurements. These are not part of the current tasks; we will consider them in the next stages of the project.
6. **The interface must be simple, especially for older users.** Martim pointed out that older patients may have difficulties with medical terms and complicated interfaces, and suggested large buttons, simple graphs, a glossary and configurable notifications, such as a subtle vibration. This supports using large and readable values, simple graphs, explaining terms such as ICR and ISF in the Profile, and allowing the user to configure how alerts are shown (Task 3).

6. Appendix 1

Interview 1: The Patient (João Antunes - Young/Adult with Type 1 Diabetes)

Interviewer: *What is the biggest challenge you face in the daily management of your diabetes?*
João Antunes: *My biggest challenge is remembering to check my glucose levels, take my medication, and maintain healthy eating habits every day. It's a continuous process that requires a lot of attention before every meal.*
Interviewer: *How could our SmartGlycoAI mobile app facilitate this management, especially on a busy day?*
João Antunes: *It would help immensely to organize my daily routine, save my glucose measurements, and remind me when I need to take my medication. If I open the app in an emergency, I need to see my current glucose level and the graph for the last few days right in the first few seconds. Having a widget on the phone's home screen would be perfect to see the blood glucose without even opening the app.*
Interviewer: *What specific features would you like the application to have, particularly regarding food?*
João Antunes: *I would like to have a glucose tracker, medication reminders, a meal log, progress graphs, and simple explanations about diabetes. Taking a photo of the plate for the app to calculate the carbohydrates would be very practical. I would feel comfortable using the camera in a restaurant, as long as the process was fast and discreet.*
Interviewer: *Would you like the application to provide personalized recommendations using Artificial Intelligence?*
João Antunes: *Yes, especially recommendations about food, exercise, and daily habits based on my glucose history. However, I would want the medical advice to be checked and validated by my healthcare team.*
Interviewer: *What would make you use the application every day?*
João Antunes: *A simple interface, a quick glucose log, useful reminders, and the ability to clearly see if my condition is improving.*

Interview 2: The Doctor (Joana Silvia - Endocrinologist)

Interviewer: *In what way could an application like SmartGlycoAI help you monitor your patients with diabetes?*
Joana Silvia: *It would help me access patients' glucose measurements, medication history, and health progress in a single place, which would make consultations much more efficient.*
Interviewer: *What features would be most important for your clinical work?*
Joana Silvia: *I would prioritize glucose history graphs, alerts for concerning patterns, organized patient records, and the ability to review medication adherence. The fact that the platform has a web version allows me to analyze this data comfortably on my office computer screen.*
Interviewer: *How could the application improve communication with your patients?*
Joana Silvia: *Patients could share their glucose measurements and symptoms between consultations, allowing me to better understand their condition and the impact of meals throughout the week.*
Interviewer: *Do you consider that AI-generated health recommendations, such as suggesting insulin doses, would be useful?*
Joana Silvia: *Yes, particularly for identifying trends and highlighting potential problems. However, AI should support medical decisions rather than replace professional judgment.*
Interviewer: *What would make you trust this application for clinical use?*
Joana Silvia: *Accurate information, secure patient records, clear explanations about the insights generated by AI, and reliable clinical data. A transparent section about the development team and how the algorithm was trained also provides greater security.*

Interview 3: The Nurse (Martim Garcia)

Interviewer: *How could this application help you in your daily follow-up work?*
Martim Garcia: *It could help me monitor patients' glucose readings, check if they are following their care routines, and easily identify those who need additional support.*
Interviewer: *What features would be most useful to ensure the success of the treatment?*
Martim Garcia: *Medication reminders, tracking patient progress, notifications about missed glucose measurements, and educational information.*
Interviewer: *How could the application help patients manage diabetes more independently?*
Martim Garcia: *It could remind them to monitor their glucose, follow their prescribed medication schedules, and maintain healthy habits.*
Interviewer: *What problems do you think patients might face when using the application, and how would you solve them?*
Martim Garcia: *Some patients might have difficulties understanding medical information or navigating complicated interfaces, particularly older users. The inclusion of a glossary for technical terms would be an excellent help.*
Interviewer: *What would you improve in the design of a diabetes management application?*
Martim Garcia: *I would make the application very easy to use, with large buttons, clear instructions, simple graphs, and notifications that patients can customize. Notifications on the phone should be configurable, such as a subtle vibration, so they are not intrusive in the workplace.*